

PART C: DATA REQUEST

To determine if STADIOalot can utilize historical data to find orders that have a higher potential for return via the analysis below are the order/transaction, product/seller, customer/customer behaviour/history of returns, as well as all types of financial and sustainability impacts data. ideally the data will include at least three or four years to allow for adequate historical observation data in addition to predictive modelling

Table 1: Order and Transaction Data

Column name	Data type	Example	Additional information
Order ID	String	ORD00001	Unique identifier for each order. Must not be duplicated or blank.
Product ID	String	PROD0318	Unique identifier for the product purchased. Must match the Product ID in the product dataset.
User ID	String	USER1469	Unique customer identifier. Must match the User ID in the customer and behavioural datasets.
Order Date	Date/DateTime	2025-03-16 14:35:21	Date and time when the order was placed. Use YYYY-MM-DD HH:MM:SS format where time is available.
Product Category	Categorical	Clothing	Standardised product category. Categories should be consistently recorded.
Product Price	Numeric, decimal	983.68	Price of one unit before discount. Must be ≥ 0 and use a consistent currency.
Order Quantity	Numeric, integer	5	Number of units purchased. Must be greater than 0.
Discount Applied	Numeric, decimal	47.80	Discount applied to the order/product. Must be ≥ 0 and use the same currency as the product price.
Shipping Method	Categorical	Express	Standardised shipping method, e.g. Standard, Express or Next-Day.
Payment Method	Categorical	Wallet	Payment method used. Valid values should be standardised, e.g. Wallet, COD, Credit Card or Debit Card.
Order Value	Numeric, decimal	2618.35	Total value of the order after applicable discounts. Must be ≥ 0 .

Column name	Data type	Example	Additional information
Delivery Distance	Numeric, decimal	125.50	Distance between fulfilment location and customer delivery location, preferably measured in kilometres.

Table 2: Product and Seller Data

Column name	Data type	Example	Additional information
Product ID	String	PROD0318	Unique identifier for each product. Must match the Product ID in the transaction data.
Seller ID	String	SELL102	Unique identifier for the seller supplying the product.
Product Category	Categorical	Clothing	Standardised product category.
Product Brand	Categorical/String	Brand A	Brand associated with the product, where applicable.
Product Size	Categorical	Medium	Size of the product. Values should be standardised according to the product category.
Product Colour	Categorical	Black	Standardised colour name.
Size Colour Combination	Categorical	Medium-Black	Combination of product size and colour/SKU.
Product Rating	Numeric, decimal	4.2	Average product rating available before the order. Preferably between 0 and 5.
Review Count	Numeric, integer	185	Number of product reviews available at the time of purchase. Must be ≥ 0 .
Product Age	Numeric, integer	180	Number of days the product has been listed on the platform at the time of purchase. Must be ≥ 0 .
Product Availability	Categorical	In Stock	Product availability at the time of purchase. Suggested values: In Stock, Low Stock or Out of Stock.

Column name	Data type	Example	Additional information
Seller Return Rate	Numeric, decimal	0.18	Historical proportion of the seller's orders that were returned, calculated using information available before the current order. Value should be between 0 and 1.

Table 3: Customer Data

Column name	Data type	Example	Additional information
User ID	String	USER1469	Unique customer identifier. Must match the User ID in the transaction data.
User Age	Numeric, integer	55	Customer age at the time of purchase. Must be a realistic non-negative value.
User Gender	Categorical	Female	Standardised categories such as Male, Female, Other or Prefer not to say.
User Location	Categorical	City5	Customer location using consistent location names/codes.
Customer Tenure	Numeric, integer	24	Number of months the customer has been registered. Must be ≥ 0 .
Previous Orders	Numeric, integer	8	Number of orders placed before the current order. Must be ≥ 0 .
Previous Returns	Numeric, integer	2	Number of previous orders returned by the customer. Must be ≥ 0 .
Previous Return Rate	Numeric, decimal	0.25	Proportion of previous orders returned. Must be between 0 and 1.
Average Order Value	Numeric, decimal	750.50	Customer's historical average order value before the current order.
Average Order Quantity	Numeric, decimal	2.4	Average number of items purchased per previous order. Must be ≥ 0 .

Table 4: Customer Behaviour and Clickstream Data

Column name	Data type	Example	Additional information
User ID	String	USER1469	Unique customer identifier. Must match the customer and transaction datasets.
Session ID	String	SES78945	Unique identifier for the customer's browsing session.
Session Date	Date/DateTime	2025-03-16 14:20:15	Date and time of the browsing session.
Product Views	Numeric, integer	8	Number of product pages viewed during the session. Must be ≥ 0 .
Search Count	Numeric, integer	4	Number of searches conducted during the session. Must be ≥ 0 .
Product Comparisons	Numeric, integer	2	Number of product comparisons made before purchase. Must be ≥ 0 .
Time on Product Page	Numeric, decimal	145.5	Time spent viewing the product, preferably measured in seconds. Must be ≥ 0 .
Previous Product Views	Numeric, integer	12	Number of times the customer previously viewed the product before purchasing. Must be ≥ 0 .
Add to Cart Count	Numeric, integer	3	Number of times the product was added to the cart. Must be ≥ 0 .
Wishlist Activity	Numeric, integer	1	Number of times the product was added to or removed from a wishlist.
Search Terms	String/Text	black size 10 shoes	Search terms used by the customer before purchase. Data should comply with applicable privacy and data-governance requirements.
Device Type	Categorical	Mobile	Device used during the session. Suggested values: Mobile, Desktop or Tablet.
Session Duration	Numeric, decimal	560.5	Total duration of the browsing session, preferably in seconds. Must be ≥ 0 .

Table 5: Historical Returns Data

Column name	Data type	Example	Additional information
Order ID	String	ORD00001	Must match an Order ID in the transaction dataset.

Column name	Data type	Example	Additional information
Return Status	Categorical	Returned	Primary target variable. Valid values should be Returned or Not Returned.
Return Reason	Categorical	Size Issue	Standardised categories such as Size Issue, Defective Product, Wrong Item, Damaged Item, Changed Mind or Other.
Return Date	Date	2025-03-28	Date on which the product was returned. Must not precede the order date.
Days to Return	Numeric, integer	12	Number of days between the order and return date. Must be ≥ 0 .
Return Quantity	Numeric, integer	1	Number of units returned. Must be greater than 0 and should not exceed the original order quantity.
Return Condition	Categorical	Damaged	Condition of the product when returned. Suggested values: New, Used, Damaged or Defective.
Refund Amount	Numeric, decimal	500.00	Amount refunded to the customer. Must be ≥ 0 .
Return Method	Categorical	Courier	Method used to return the product, e.g. Courier, Collection or Drop-off.
Return Processing Time	Numeric, decimal	3.5	Number of days required to process the return. Must be ≥ 0 .

Table 6: Financial and Sustainability Impact Data

Column name	Data type	Example	Additional information
Order ID	String	ORD00001	Must match the Order ID in the transaction dataset.
Return Cost	Numeric, decimal	200.00	Total cost incurred in processing the return. Must be ≥ 0 .
Profit/Loss	Numeric, decimal	2418.35	Financial outcome associated with the order after relevant costs. Use a consistent currency.
CO2 Emissions	Numeric, decimal	2.00	Carbon emissions associated with fulfilment. Specify the unit, preferably kg CO ₂ e.
Packaging Waste	Numeric, decimal	1.00	Amount of packaging material used. Specify the unit, preferably kilograms.

Column name	Data type	Example	Additional information
CO2 Saved	Numeric, decimal	2.00	Estimated CO ₂ emissions avoided through sustainability activities. Specify the calculation methodology and unit.
Waste Avoided	Numeric, decimal	1.00	Estimated quantity of waste avoided. Specify the unit and calculation methodology.
Return Shipping Distance	Numeric, decimal	130.20	Distance travelled to transport the returned product, preferably in kilometres.
Return Emissions	Numeric, decimal	1.20	CO ₂ emissions generated by return transportation. Specify the unit as kg CO ₂ e.
Disposal Cost	Numeric, decimal	150.00	Cost of disposing of returned products that cannot be resold. Must be ≥ 0 .
Resale Status	Categorical	Resold	Outcome of the returned product. Suggested values: Resold, Refurbished, Recycled or Disposed.